

MH2500: Probability and Intro to Statistics

Crash Course Revision

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- **Conditional Probability**

$$P(A | B) = \frac{P(A \cap B)}{P(B)}.$$

- **Multiplicative rule**

$$P(A \cap B) = P(A | B) \cdot P(B).$$

- **Inclusion-Exclusion Principle**

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

- **Law of Total Probability (LOTP)**

$$P(A) = P(A | B_1)P(B_1) + P(A | B_2)P(B_2) + \cdots + P(A | B_n)P(B_n).$$

where B_i 's are all mutually disjoint and all the B_i 's form the sample space Ω .

- **Bayes' Rule**

(‘Cause given Consequence’ $\rightarrow$ ‘Consequence given Cause’)

$$P(B_j | A) = \frac{P(A | B_j)P(B_j)}{\sum_{i=1}^n P(A | B_i)P(B_i)} = \frac{P(A | B_j)P(B_j)}{P(A)}$$

Example: A healthy study tracked a group of persons for five years. During the study period, 20% of the persons were classified as heavy smokers, 30% as light smokers and 50% as non-smokers. Results of the study showed that light smokers were twice as likely as nonsmokers to die during the five-year study, but only half as likely as heavy smokers.

A randomly selected participant from the study died over the five-year period. Calculate the probability that the participant was a heavy smoker.

- **Independent events**

$P(A \cap B) = P(A)P(B)$ implies A and B are independent.

- **CDF of Discrete Random Variables:** $F(x) = P(X \leq x)$.
 - $P\{a < X \leq b\} = P\{X \leq b\} - P\{X \leq a\} = F(b) - F(a)$.
 - $P\{X < a\} = \lim_{x \rightarrow a^-} F(x)$.
 - $P\{a \leq X \leq b\} = P\{X \leq b\} - P\{X < a\} = F(b) - \lim_{x \rightarrow a^-} F(x)$.
 - $P(X = a) = P(X \leq a) - P(X < a) = F(a) - \lim_{x \rightarrow a^-} F(x)$.
- **Expectation and Variance**

$$\mu_X = \mathbb{E}(X) = \sum_i x_i p(x_i), \quad \text{Var}(X) = \mathbb{E}(X^2) - \{\mathbb{E}(X)\}^2.$$

- **Poisson approximation of Binomial r.v.** Given $Y \sim \text{Bin}(n, p)$ we can use $Y \sim \text{Poisson}(np)$ approximately.

$$\mathbb{E}(Y) = np \rightarrow \lambda, \quad \text{Var}(Y) = np(1-p) \rightarrow \lambda$$

when $n \rightarrow \infty$.

(Remark: This approximation is good if $n \geq 50$ and $np \leq 5$.)

Distribution	pmf $P(X = k)$	$\mathbb{E}(X)$	$\text{Var}(X)$
Indicator/ Bernoulli(p)	$\begin{cases} p & \text{if } k \in \text{Event} \\ 1 - p & \text{if } k \notin \text{Event} \end{cases}$	p	$p(1 - p)$
Binomial(n, p)	$\binom{n}{k} p^k (1 - p)^{n-k}$	np	$np(1 - p)$
Poisson(λ)	$e^{-\lambda} \frac{\lambda^k}{k!}$	λ	λ
Geometric(p)	$(1 - p)^{k-1} p$	$\frac{1}{p}$	$\frac{1 - p}{p^2}$
Hypergeometric (n, N, m) pick k/n 'whites'	$\frac{\binom{m}{k} \binom{N-m}{n-k}}{\binom{N}{n}}$	$n \frac{m}{N}$	$n \frac{m}{N} \left(1 - \frac{m}{N}\right) \left(\frac{N-n}{N-1}\right)$

You may use $\mathbb{E}(X^2) = \text{Var}(X) + \{\mathbb{E}(X)\}^2$ to obtain $\mathbb{E}(X^2)$.

- **Properties of pdf $f(x)$ and cdf $F(x)$.**

$f(x)$ is a pdf if $f(x) \geq 0$ for all $x \in \mathbb{R}$ and $\int_{-\infty}^{\infty} f(x) dx = 1$.

$$f(x) = \frac{d}{dx}[F(x)] \text{ and } P(a \leq X \leq b) = F(b) - F(a).$$

- **Expectation and Variance**

$$\mu_X = \mathbb{E}(X) = \int_{-\infty}^{\infty} xf(x) dx, \quad \mathbb{E}(X^2) = \int_{-\infty}^{\infty} x^2 f(x) dx, \quad \text{Var}(X) = \mathbb{E}(X^2) - \{\mathbb{E}(X)\}^2.$$

- **Standardization:** use of $Z = \frac{X-\mu}{\sigma} \sim N(0,1)$. Standard Tables

$$P(Z \leq a) = \Phi(a) = 1 - \Phi(-a) \text{ for any } a \in \mathbb{R}.$$

- **Normal Approximation to Binomial Distribution:** approximation is good when $np(1-p) \geq 10$.
To approximate $X \sim \text{Bin}(n, p)$, apply standardization and continuity correction. (Draw number line to help).

Distribution	density / pdf	cdf $P(X \leq x)$	$\mathbb{E}(X)$	$\text{Var}(X)$
Uniform(a, b)	$\frac{1}{b-a}$ $a < x < b$	$\frac{x-a}{b-a}$ $a < x < b$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Exponential(λ)	$\lambda e^{-\lambda x}, x \geq 0$	$1 - e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Normal(μ, σ^2)	$\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	$\frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^x e^{-\frac{(t-\mu)^2}{2\sigma^2}} dt$	μ	σ^2
$Z \sim (0, 1)$	$\frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$	Tables	μ	σ^2
Gamma(α, λ)	$\frac{\lambda^\alpha}{\Gamma(\alpha)} t^{\alpha-1} e^{-\lambda t},$ $t \geq 0$ $\Gamma(n) = (n-1)!$	$\frac{\lambda^\alpha}{\Gamma(\alpha)} \int_0^x t^{\alpha-1} e^{-\lambda t} dt$	$\frac{\alpha}{\lambda}$	$\frac{\alpha}{\lambda^2}$

- **pdf/cdf of function of Random variable**

Suppose X is a random variable with density function $f_X(x)$ and Y is a random variable that can be expressed in terms of X , say, $Y = g(X)$ for some function g .

How to determine density function of Y (or $f_Y(y)$)?

- 1 Find the CDF of Y (or $F_Y(y)$), i.e., $P\{Y \leq y\}$.
(Express in terms of probability of event about X)
In other words, obtain $P(X \leq h(y))$.
 - Method 1) Find the cdf of Y using integration.
 - Method 2) Keep it as $F(h(y))$.
- 2 Then, differentiate the CDF with respect to y .
*Note: Remember to put the range of values of $y = g(x)$ can take based on the range of values x can take.

(Weeks 8-9) Joint Distributions

#Note: I only included Joint continuous r.v., DIY for discrete case.

- **Joint cdf $F(x, y)$ and pdf $f(x, y)$**

Conditions for pdf: $f(x, y) \geq 0$, $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dy dx = 1$.

Joint CDF: $F(x, y) = P\{X \leq x, Y \leq y\} = \int_{-\infty}^x \int_{-\infty}^y f(u, v) dv du$.

- **Marginal pdf of X and Y** (Based on region of $f(x, y)$)

$$f_X(x) = \underbrace{\int_{-\infty}^{\infty} f(x, y) dy}_{\text{Integrate over all the } y} \quad \text{and} \quad f_Y(y) = \underbrace{\int_{-\infty}^{\infty} f(x, y) dx}_{\text{Integrate over all the } x}.$$

- **Independence of X and Y .**

Check $f_{XY}(x, y) = f_X(x)f_Y(y)$ for all x, y . Else, not independent.

- **Conditional distribution of Y given X .**

$$f_{Y|X}(y|x) = \begin{cases} \frac{f_{XY}(x, y)}{f_X(x)}, & \text{if } f_X(x) > 0 \\ 0, & \text{otherwise.} \end{cases}$$

• Sum of Random Variables (Convolution)

- For continuous random variables X and Y , $Z = X + Y$ has the pdf:

$$f_Z(z) = \int_{-\infty}^{\infty} f_{XY}(x, z-x) dx.$$

- If X and Y are **independent**, $Z = X + Y$ has the pdf:

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx.$$

• Ordered Statistics

Given $X_1, X_2, \dots, X_n$ are **i.i.d** random variables with cdf F and pdf f .
Let U be the **maximum** of X_i 's and V be the **minimum** of X_i 's.

- $P(U \leq u) = (F(u))^n$ and $f_U(u) = n(F(u))^{n-1} f(u)$.
- $P(V \leq v) = 1 - (1 - F(v))^n$ and $f_V(v) = n(1 - F(v))^{n-1} f(v)$.

• Joint Distribution of Max and Min

$$f(u, v) = \begin{cases} 0, & \text{if } u \leq v; \\ n(n-1)f(u)f(v)(F(u) - F(v))^{n-2}, & \text{if } u \geq v. \end{cases}$$

If X_i 's are uniform on $[0, 1]$, then

$$f(u, v) = n(n-1)(u-v)^{n-2} \quad 1 \geq u \geq v \geq 0.$$

- Expectations of function of Joint r.v.**

If X and Y have a joint pmf $p(x, y)$ or pdf $f(x, y)$ then

$$\mathbb{E}(g(X, Y)) = \sum_{x, y} g(x, y)p(x, y) \text{ or } \iint_{\mathbb{R}^2} g(x, y)f(x, y) dx dy.$$

where $\mathbb{R}^2$ is the region where the joint pdf is defined.

Examples:

$$\mathbb{E}(X) = \iint_{\mathbb{R}^2} x f(x, y) dx dy, \quad \mathbb{E}(XY) = \iint_{\mathbb{R}^2} xy f(x, y) dx dy.$$

- Covariance.**

$$Cov(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y).$$

If X and Y are independent, then $Cov(X, Y) = 0$. Converse is false.

If $X_1, X_2, \dots, X_n$ are mutually independent, then

$$Var(X_1 + X_2 + \dots + X_n) = Var(X_1) + Var(X_2) + \dots + Var(X_n).$$

● Results from Covariance.

- $Cov(a, Y) = 0$.
- $Cov(X, Y) = Cov(Y, X)$.
- $Cov(X, X) = \mathbb{E}(X^2) - \mathbb{E}(X) \cdot \mathbb{E}(X) = Var(X)$.
- $Cov(aX, bY) = ab \cdot Cov(X, Y)$.
- $Cov(aX + bY, Z) = a \cdot Cov(X, Z) + b \cdot Cov(Y, Z)$.
- $Cov(aW + bX, cY + dZ) = ac \cdot Cov(W, Y) + ad \cdot Cov(W, Z) + bc \cdot Cov(X, Y) + bd \cdot Cov(X, Z)$.
- $Var(X + Y) = Var(X) + Var(Y) + 2 \cdot Cov(X, Y)$ for any X, Y .
- $Var\left(a + \sum_{i=1}^n b_i X_i\right) = \sum_{i=1}^n b_i^2 Var(X_i) + 2 \sum_{j=i+1}^n \sum_{i=1}^n b_i b_j Cov(X_i, X_j)$.

● Correlation Coefficient ρ_{XY} .

$$\rho_{XY} = \frac{Cov(X, Y)}{\sqrt{Var(X)Var(Y)}}, \quad -1 \leq \rho \leq 1.$$

● Conditional Expectation $\mathbb{E}(g(Y) \mid X = x)$.

$$\text{Discrete: } \mathbb{E}(g(Y) \mid X = x) = \sum_y g(y) p_{Y|X}(y|x).$$

$$\text{Continuous: } \mathbb{E}(g(Y) \mid X = x) = \int_y g(y) f_{Y|X}(y|x) dy.$$

- **Law of Total Expectation.**

Mean of $\mathbb{E}(Y | X)$ is $\mathbb{E}[\mathbb{E}(Y | X)]$. Note: $\mathbb{E}(Y) = \mathbb{E}_X[\mathbb{E}_Y(Y | X)]$.

Discrete: $\mathbb{E}(Y) = \sum_x \mathbb{E}(Y | X = x)p_X(x)$. Continuous: $\mathbb{E}(Y) = \int_x \mathbb{E}(Y | X = x)f_X(x) dx$.

- **Conditional Variance.**

For any value of x , we define

$$\text{Var}(Y|X = x) = \mathbb{E}(Y^2|X = x) - [\mathbb{E}(Y|X = x)]^2.$$

'Interchangeable' formula: $\text{Var}(Y) = \text{Var}[\mathbb{E}(Y|X)] + \mathbb{E}[\text{Var}(Y|X)]$.

- **Indicator random variables by conditioning.** Given that E is an event:

$$P(E) = \sum_x P(E|X = x)p(x) \text{ or } \int_x P(E|X = x)p(x)dx.$$

- **Coupon Collector Problem:** Given N distinct coupon types, with each type equally likely to be chosen, the expected number of coupons needed to collect the complete set of N coupons at least once is,

$$\mathbb{E}(X) = 1 + \frac{1}{(N-1)/N} + \frac{1}{(N-2)/N} + \cdots + \frac{1}{2/N} + \frac{1}{1/N} = N \left(1 + \frac{1}{2} + \cdots + \frac{1}{N-1} + \frac{1}{N} \right).$$

- **Markov's Inequality:** Let X be a *non-negative* random variable, then

$$P\{X \geq t\} \leq \frac{\mathbb{E}(X)}{t} \quad \text{for any } t > 0.$$

- **Chebyshev's Inequality:** Let X be a random variable such that $\mathbb{E}(X)$ and $\text{Var}(X)$ exist. Then,

$$P\{|X - \mathbb{E}(X)| \geq t\} \leq \frac{\text{Var}(X)}{t^2} \quad \text{for any } t > 0.$$

- **Sample Mean, $\bar{X}_n$, and Sum, S_n , of i.i.d random variables.**

Let $X_1, X_2, \dots, X_n$, be i.i.d. variables with mean μ and variance σ^2 .

Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$, $S_n = \sum_{i=1}^n X_i$. Then,

$$\mathbb{E}(\bar{X}_n) = \mu, \text{Var}(\bar{X}_n) = \frac{\sigma^2}{n} \text{ and } \mathbb{E}(S_n) = n\mu, \text{Var}(S_n) = n\sigma^2.$$

- **Central Limit Theorem.** Let $X_1, X_2, \dots$ be a sequence of **i.i.d.** r.v. having mean μ and variance σ^2 . Then,

$$\lim_{n \rightarrow \infty} P\left\{ \frac{S_n - n\mu}{\sigma\sqrt{n}} \leq x \right\} = \Phi(x), \quad -\infty < x < \infty.$$

Apply continuity correction if X_i happens to be discrete.